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Binary Diffusion Probabilistic Models

Computational Imaging
14x062



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1 Introduction on Diffusion Models

Generative models have revolutionized the field of artificial intelligence, enabling machines to create content that is indistinguishable from human-generated data or to improve the quality of data through various transformations. Among these models, diffusion models have emerged as a powerful class of generative models that have shown remarkable performance in tasks such as image generation, image super-resolution, and data denoising.

Diffusion models are based on the concept of a diffusion process, which is a stochastic process that describes how data points evolve over time. The core idea behind diffusion models is to model the data generation process as a gradual transformation from noise to structured data. This is achieved by defining a forward diffusion process that adds noise to the data and a reverse diffusion process that learns to remove the noise and recover the original data.

Among the various types of diffusion models, Denoising Diffusion Probabilistic Models (DDPMs) have gained significant attention due to their ability to generate high-quality samples. DDPMs work by iteratively adding noise to the data and then learning to reverse this process, effectively denoising the data step by step.

But are these models very suitable for all types of data, especially for quantized data? Quantized data, which is often used in scenarios like in computational imaging, presents unique challenges for generative modeling. The discrete nature of quantized data can make it difficult for diffusion models to learn the underlying data distribution effectively.

In answer to this problem, researchers have proposed various approaches to adapt diffusion models for quantized data, such as Binary Diffusion Models (BDMs) and Quantized Diffusion Models (QDMs). These models are designed to handle the discrete nature of quantized data while still leveraging the powerful generative capabilities of diffusion models.

2 Methodology

There are several types of generative models, including:

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)
- Normalizing Flows
- Diffusion Models

Each of these models has its own strengths and weaknesses, and the choice of model often depends on the specific application and the type of data being modeled.

Diffusion models, in particular, have shown great promise in generating high-quality samples and have been successfully applied to a wide range of tasks. However, their performance can be limited when dealing with quantized data, which is often encountered in scenarios like computational imaging. To address this issue, researchers have developed specialized diffusion models that are designed to handle quantized data effectively.

First, what are the foundational principles of diffusion models?

Diffusion models are a Markov chain-based generative model that defines a forward diffusion process and a reverse diffusion process. The forward diffusion process gradually adds noise to the data, while the reverse diffusion process learns to remove the noise and recover the original data. The training of diffusion models involves optimizing a variational bound on the data likelihood, which allows the model to learn the underlying data distribution effectively.

2.1 Forward Diffusion Process

Suppose we have an original image X_0 that follows a data distribution q .

In Denoising Diffusion Probabilistic Models (DDPMs), the forward diffusion process is defined as first order Markov chain that gradually adds noise to the data over a series of time steps.

At each time step t , noise is added to the data according to a predefined schedule, which is typically controlled by a parameter β_t that determines the amount of noise added at each step, as shown in Equation 1 where $\varepsilon_t \sim \mathcal{N}(0, I)$ is a standard Gaussian noise term and X_t is the noisy version of the data at time step t .

$$X_t = \sqrt{1 - \beta_t} X_{t-1} + \sqrt{\beta_t} \varepsilon_t \quad (1)$$

Indeed, if $\beta_t = 1$ in Equation 1, then X_t would be pure noise whereas if $\beta_t = 0$, then X_t would be the original data X_{t-1} .

This process can be visualized as a gradual transformation of the original data into noise, where the data becomes increasingly corrupted as more noise is added at each time step, as illustrated in the Figure 1.

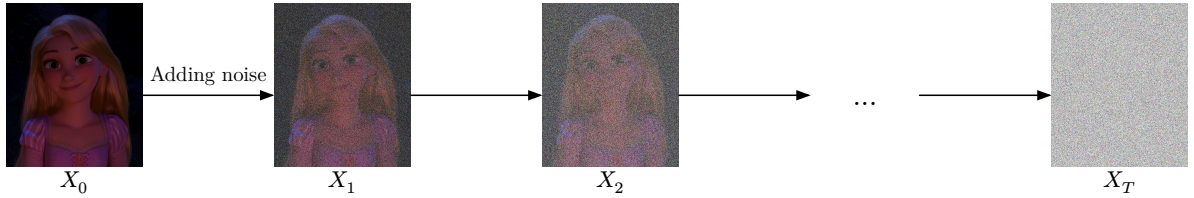


Figure 1: Illustration of the forward diffusion process

We can also express X_t in term of the data distribution q as described in Equation 2, where $\mathcal{N}(X; \mu, \sigma^2)$ denotes a Gaussian distribution with mean μ and variance σ^2 applied to X .

$$q(X_t | X_{t-1}) = \mathcal{N}(X_t; \sqrt{1 - \beta_t} X_{t-1}, \beta_t I) \quad (2)$$

By iterating this process over T time steps, we can obtain a sequence of noisy data points $(X_1, X_2, \dots, X_T)$, where X_T is the noisiest version of the data which can be considered as pure noise.

This can be further simplified to a closed-form expression that directly relates X_T to the original data X_0 , as shown in Equation 3.

$$q(X_T | X_0) = q(X_1 | X_0) \times q(X_2 | X_1) \times \dots \times q(X_T | X_{T-1}) = \prod_{t=1}^T q(X_t | X_{t-1}) \quad (3)$$

We note $\alpha_t = 1 - \beta_t$, so the forward process can be written as:



$$\begin{aligned}
 X_t &= \sqrt{\alpha_t} X_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_t \\
 &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} X_{t-2} + \sqrt{1 - \alpha_{t-1}} \varepsilon_{t-1}) + \sqrt{1 - \alpha_t} \varepsilon_t \\
 &= \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} X_{t-2} + \sqrt{\alpha_t} \sqrt{1 - \alpha_{t-1}} \varepsilon_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_t \\
 &= \sqrt{\alpha_t \alpha_{t-1}} X_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_t \\
 &= \sqrt{\alpha_t \alpha_{t-1}} (\sqrt{\alpha_{t-2}} X_{t-3} + \sqrt{1 - \alpha_{t-2}} \varepsilon_{t-2}) + \sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_t \\
 &= \sqrt{\alpha_t \alpha_{t-1} \alpha_{t-2}} X_{t-3} + \sqrt{\alpha_t \alpha_{t-1} (1 - \alpha_{t-2})} \varepsilon_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-1} + \sqrt{1 - \alpha_t} \varepsilon_t \quad (4) \\
 &= \dots \\
 &= \sqrt{\prod_{i=1}^t \alpha_i} X_0 + \sum_{k=1}^t \sqrt{(1 - \alpha_k) \prod_{j=k+1}^t \alpha_j} \varepsilon_k \\
 &= \sqrt{\bar{\alpha}_t} X_0 + \sum_{k=1}^t \sqrt{(1 - \alpha_k) \prod_{j=k+1}^t \alpha_j} \varepsilon_k \quad \text{with } \bar{\alpha}_t = \prod_{i=1}^t \alpha_i
 \end{aligned}$$

We note $N_t = \sum_{k=1}^t \sqrt{(1 - \alpha_k) \prod_{j=k+1}^t \alpha_j} \varepsilon_k$. We want to express N_t as a single noise term since a sum of independent Gaussian noise terms is also a Gaussian noise term with variance equal to the sum of the variances of the individual noise terms. The variance of N_t can be calculated as in Equation 5, where we use the fact that $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_T \sim \mathcal{N}(0, I)$ are independent standard Gaussian noise terms.

$$\begin{aligned}
 \text{Var}(N_t) &= \text{Var} \left(\sum_{k=1}^t \sqrt{(1 - \alpha_k) \prod_{j=k+1}^t \alpha_j} \varepsilon_k \right) \\
 &= \sum_{k=1}^t (1 - \alpha_k) \prod_{j=k+1}^t \alpha_j \text{Var}(\mathcal{N}(0, I)) \\
 &= \sum_{k=1}^t (1 - \alpha_k) \prod_{j=k+1}^t \alpha_j \\
 &= (1 - \alpha_t) + \alpha_t (1 - \alpha_{t-1}) + \alpha_t \alpha_{t-1} (1 - \alpha_{t-2}) + \dots + \prod_{i=2}^t \alpha_i (1 - \alpha_1) \quad (5) \\
 &= 1 - \alpha_t + \alpha_t - \alpha_t \alpha_{t-1} + \alpha_t \alpha_{t-1} - \alpha_t \alpha_{t-1} \alpha_{t-2} + \dots + \prod_{i=2}^t \alpha_i - \prod_{i=1}^t \alpha_i \\
 &= 1 - \prod_{i=1}^t \alpha_i \\
 &= 1 - \bar{\alpha}_t
 \end{aligned}$$

We obtain the final expression for X_t in Equation 6 with $\varepsilon \sim \mathcal{N}(0, I)$ being a single Gaussian noise term with variance $1 - \bar{\alpha}_t$.

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon \quad (6)$$

This closed-form expression is particularly useful because it allows us to sample X_t directly from X_0 without having to iterate through all the intermediate steps, which can be computationally expensive when T is large.

2.2 Reverse Diffusion Process

The reverse diffusion process consists of going from the noisy data back to the original data by learning to reverse the noise addition process, as illustrated in Figure 2.

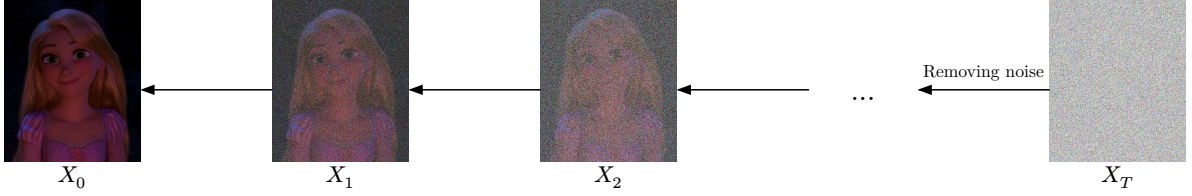


Figure 2: Illustration of the reverse diffusion process

As we cannot estimate $q(X_{t-1} | X_t)$ directly, we train a model $p_\theta(X_{t-1} | X_t)$ to approximate it, where θ denotes the parameters of the model. This conditional distribution can be represented as a Gaussian distribution with mean $\mu_\theta(X_t, t)$ and covariance $\Sigma_\theta(X_t, t)$, as shown in Equation 7.

$$p_\theta(X_{t-1} | X_t) = \mathcal{N}(X_{t-1}; \mu_\theta(X_t, t), \Sigma_\theta(X_t, t)) \approx q(X_{t-1} | X_t) \quad (7)$$

The $\mu_\theta(X_t, t)$ aims to predict the expected value of X_{t-1} given X_t , while $\Sigma_\theta(X_t, t)$ captures the uncertainty in this prediction.

As detailed in the very nice blog post by Lilian Weng on diffusion models [1], the mean $\mu_\theta(X_t, t)$ can be expressed in terms of the noise prediction $\varepsilon_\theta(X_t, t)$ as shown in Equation 8, where α_t and $\bar{\alpha}_t$ are defined as in the forward diffusion process. In fact, the model aims to predict the noise $\varepsilon_\theta(X_t, t)$ that was added to the data at time step t in the forward diffusion process, and then uses this prediction to compute the mean $\mu_\theta(X_t, t)$ for the reverse diffusion process.

$$\mu_\theta(X_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(X_t, t) \right) \quad (8)$$

A common choice for the covariance $\Sigma_\theta(X_t, t)$ is to set it to a fixed value, such as $\Sigma_\theta(X_t, t) = \beta_t I$, which corresponds to the variance of the noise added in the forward diffusion process at time step t . Indeed, the covariance will indicate how much noise should be removed at each step of the reverse diffusion process, that's why it is often set to a value that matches the noise level in the forward diffusion process to ensure that the model learns to effectively reverse the noise addition process.

2.2.1 Intuitive interpretation

We can imagine that the data are represented by a mountain with X_0 at the peak. The forward diffusion process can be seen as a hiker starting at the peak and gradually descending the mountain, adding noise to the data at each step. The reverse diffusion process can be thought of as a hiker starting at the base of the mountain (the noisiest data) and trying to climb back up to the peak (the original data) by following the path of least resistance, which is determined by the learned model p_θ .

The $\mu_\theta(X_t, t)$ indicates the direction in which the hiker should move to climb back up the mountain, while $\Sigma_\theta(X_t, t)$ represents the uncertainty in this direction, which indicates how many different paths the hiker can take to reach the peak.

So if μ_θ is zero, the hiker don't know which direction to go and can take any path, while if μ_θ is large, the hiker has a clear direction to follow. And if Σ_θ is small, the hiker has only one path to follow, while if Σ_θ is large, the hiker has many different paths to choose from.

For instance, in the case of image generation, μ_θ indicates that the image should be a bear, while Σ_θ indicates if it should be a black bear, a brown bear, or a polar bear.

So the model aims to minimize the difference between the predicted distribution $p_\theta(X_{t-1} | X_t)$ and the true distribution $q(X_{t-1} | X_t)$ to learn how to effectively reverse the noise addition process and recover the original data from the noisy data.

The complete process can be represented as a Markov chain that alternates between the forward diffusion process and the reverse diffusion process, where the model learns to effectively reverse the noise addition process at each step to recover the original data from the noisy data, as illustrated in Figure 3.

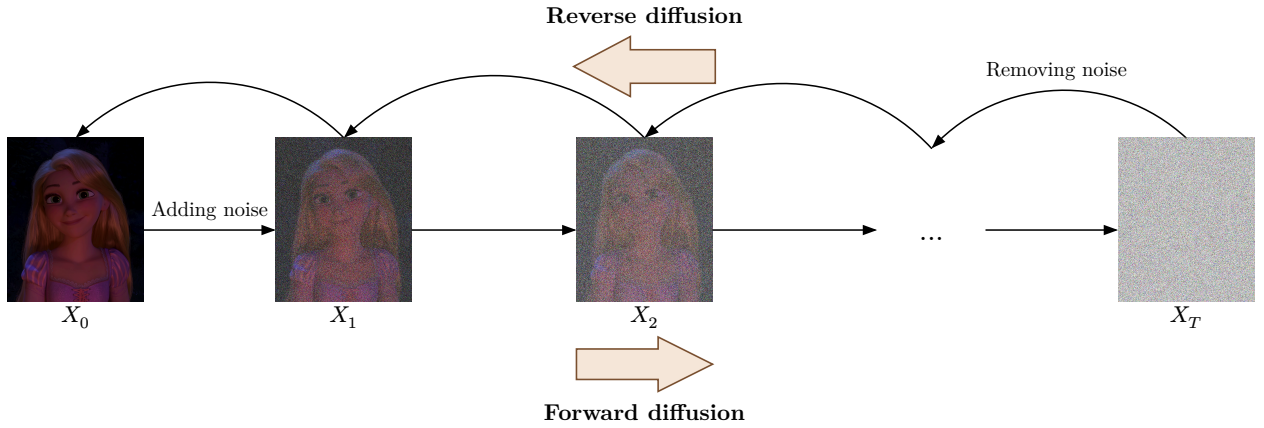


Figure 3: Illustration of the complete diffusion process, which consists of a forward diffusion process that adds noise to the data and a reverse diffusion process that learns to remove the noise and recover the original data

3 Binary Diffusion Models

The binary diffusion models are a specific type of diffusion models that are designed to handle binary data, which is a common type of quantized data. For instance, in the context of computational imaging, we can deal with binary representations of images by a decomposition of the image into bit-planes, where each bit-plane represents a binary image that captures a specific bit of the pixel values.

In binary diffusion models, thanks to the binary nature of the data, the management of the noise addition and removal process is simplified compared to the general case of continuous data. The forward diffusion process can be defined as a simple XOR operation between the binary data and a binary noise term, while the reverse diffusion process can be defined as a simple XNOR operation between the noisy data and a predicted binary noise term.

Thanks to the binary nature of the data, there is no need to tune a complex noise schedule as in the case of continuous data. This allows for a simpler preprocessing of the data and a more



straightforward training of the model, which can lead to faster convergence and improved performance when dealing with binary data.

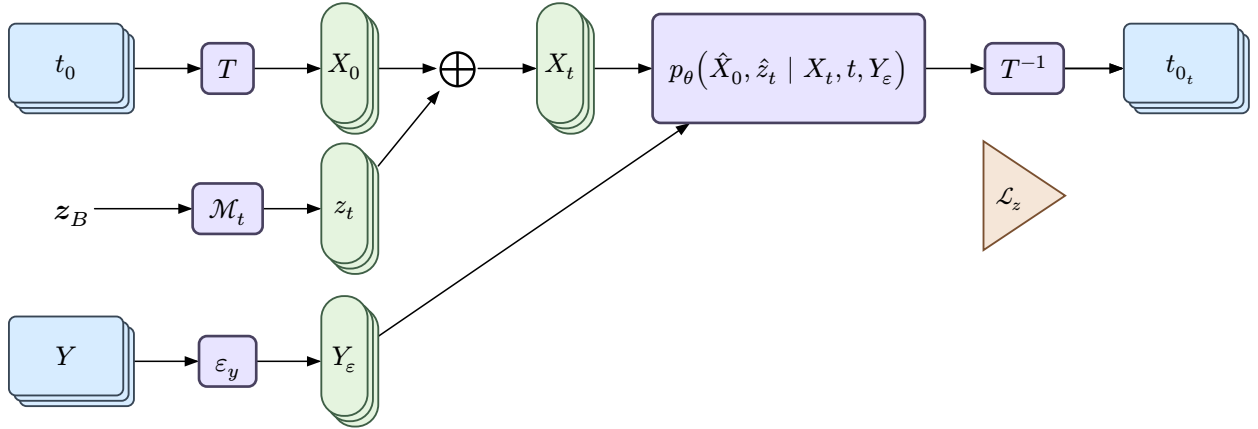


Figure 4: Binary diffusion process



Bibliography

- [1] L. Weng, “What are diffusion models?,” *lilianweng.github.io*, July 2021, [Online]. Available: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>